

Comparative Analysis of Classification Techniques in Data Mining: K-Nearest Neighbors, Naive Bayes, and Decision Trees based Approach

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Abstract: The world is running on rapid digital transformation right now; the explosion of data brings immense opportunities and some serious challenges for increasing the value of the information. Data mining is not just about getting success on picking an algorithm. It is about finding the right method with the most convenient computing environment and the details of a specific field. There are three classification techniques which are mostly used techniques in the world right now. K-Nearest Neighbors is often praised and used for the straightforward design and the ability to adapt localized data in the most useful way. But there are also some drawbacks of this technique, it struggles with scaling issues and can become unreliable in the large set of data. Naïve Bayes remains a go-to method for probability calculation, it is mainly valued for the speed it has and the surprisingly strong performance in areas such as text classification, biomedical data, and genetic studies like the cases where its assumption for featuring independence holds up well. Decision Trees such as J48 using the Gini Index bring something incompatible to the table like a simple, rule-based structure that is easy to illuminate. This transparency makes them reliable in domains like medical bioinformatics, where trust and clarity in decision-making are most crucial part. This is not only about algorithms. The platforms used for applying them to play a major role in their own effectiveness. The platforms are praised for their accessible and workflow-oriented design. These ecosystems make it easier to connect the algorithms with complex datasets, such as leukemia gene-expression profiles. The difference between the approaches are accuracy, efficiency and interpretability. They give different accuracy with different training times. The most noticeable thing is no single approach comes with everything in them.

Keywords: Supervised Learning, k-Nearest Neighbors, Naive Bayes, Decision Trees (J48/Gini), Data Mining Tools, Comparative Analysis.

1. INTRODUCTION

The rapid growth in digital data within modern society makes data mining one of the most important tool for extracting useful information from large and complex data. Among the various tasks in data mining, classification is one of the key activities that enable the grouping of data instances into predefined classes based on discovered patterns learned from labeled data [1], [2]. Applications of classification techniques include health diagnosis, fraud detection, spam filtering, sentiment analysis, prediction of customer churn, risk assessment, etc., within domains of health care, finance, e-commerce, transportation, cybersecurity, and social media analytics [3]–[5].

Despite the recent interest in deep learning and ensemble-based methods, classical classifiers include K-Nearest Neighbors (KNN), Naïve Bayes (NB), and Decision Trees (DT), which are of high relevance owing to their simplicity, interpretability, and effectiveness in a wide range of real-world settings [6]–[8]. KNN is an instance-based, non-parametric algorithm that assigns labels to new instances based on the majority class among their closest neighbors in feature space [6], [7]. Naïve Bayes is a probabilistic classifier which assumes conditional independence among features and applies Bayes' theorem to predict the most likely class. Even though this is generally an unrealistic independence assumption, NB is computationally efficient and performs well on high-dimensional and sparse datasets, such as classification of text or documents [8]–[10]. Decision Trees are

learned via recursive partitioning, where a hierarchy of if-then-else rules is created, hence making them highly interpretable and applicable to domains where decisions should be transparent [11]–[14].

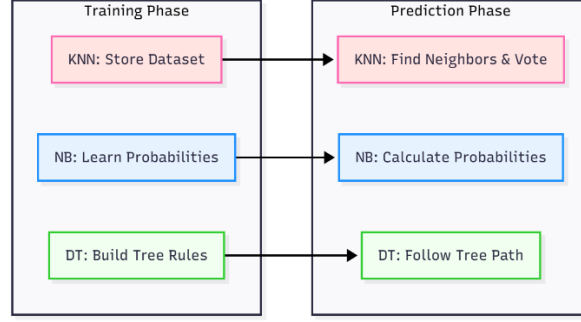


Figure I.I: Training vs. Prediction Workflow of KNN, Naïve Bayes, and Decision Tree Classifiers

Choosing the most suitable algorithm for classification is no easy task, as each has unique merits and defects. K-nearest neighbor approaches are intuitive and often yield high accuracy on small to medium datasets but suffer from the high computational cost during prediction and sensitivity to noisy or irrelevant features [6], [15], [16]. Naive Bayes is very fast to train and predict, with good robustness to irrelevant features and doing well in high-dimensional data; poor performance may be seen when there are strong feature-feature correlations [8], [9], [17]. Decision Trees are flexible and interpretable but tend to overfit the training data and are sensitive to small variations in the input of such a nature as to create significantly different tree topologies [11], [18], [19].

Comparison studies are essential to understand the practical performance of these classifiers. Although many works have benchmarked KNN, NB, and DT on the standard datasets, most of them focus on accuracy only and do not consider other important aspects such as robustness, scalability, interpretability, and performance on high-dimensional or imbalanced data [15], [20], [21]. Moreover, most contemporary applications require a model that is not only accurate but also transparent and computationally viable in real-time or resource-constrained settings [22], [23].

The purpose of this work is to present an overall comparative study between the KNN, Naive Bayes, and Decision Tree classifiers. This work shows the performance evaluation of these models in terms of several datasets and various performance metrics like accuracy, precision, recall, F1-score, and computational efficiency. It also concentrates on aspects like interpretability, sensitivity to hyperparameters, and suitability for datasets with different characteristics. Thus, this research will provide helpful recommendations for data scientists and domain practitioners in choosing an appropriate classifier for their specific application [24]–[28].

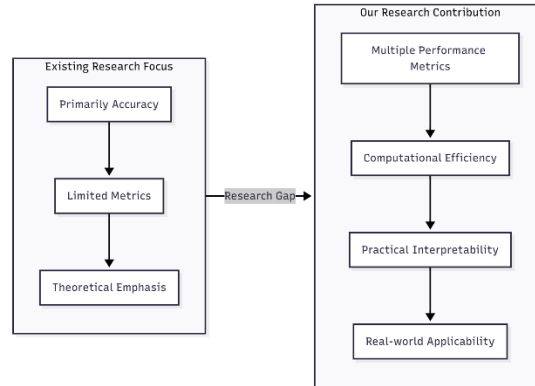


Figure I.II: Research Gap Between Existing Studies and Proposed Contribution

This paper contributes to the literature by:

1. A holistic empirical comparison using several real-world datasets on KNN, NB, and DT.
2. Discuss predictive performance and computational cost in terms of trade-off among accuracy, speed, and interpretability.
3. Discussion of the appropriateness of each classifier concerning different dataset characteristics like dimensionality, feature correlation, and class imbalance.
4. Incorporating insights from prior studies while addressing existing research gaps in classifier comparison [29], [31].

The rest of this paper is organized as follows: Section II presents the detailed literature review on KNN, Naïve Bayes, and Decision Trees, incorporating variants and applications. Section III outlines the methodology and experimental setup. The results and comparative analysis will be presented in Section IV, followed by a discussion in Section V. Section VI concludes with the key findings and future research directions.

2. LITERATURE REVIEW

2.1 Classification in Data Mining

Classification is a type of supervised learning that predicts categorical labels for any given unknown instance using a learned model [1], [2]. A variety of textbooks and surveys emphasize that the main evaluation criteria for classifiers are their accuracy, computational efficiency, interpretability, and robustness to noise [3], [4], [11], [12]. More recent studies confirm that no classifier is consistently superior to others across all data sets, as reflected by the "No Free Lunch" theorem [32]. Comparisons, therefore, remain an important research issue for methodological developments and practical applications [15], [18], [33].

2.2 K-Nearest Neighbors (KNN)

Introduced by Cover and Hart [6], KNN is a non-parametric algorithm that classifies instances based on the majority label of their nearest neighbors, typically measured using Euclidean or Manhattan distance. The algorithm, due to its simplicity and capability of modelling complex decision boundaries, is widely used in pattern recognition and medical diagnosis problems [6], [7], [16]. The advantages of KNN include high flexibility, adaptability to nonlinear decision boundaries, and conceptual simplicity [15], [21].

However, it suffers from high computational cost at the time of prediction, memory dependency on training data, sensitivity regarding noisy features, and the need to choose an appropriate value of k [7], [16], [24]. Improvements such as weighted KNN and metric learning have been suggested in the literature to enhance predictive performance and reduce sensitivity due to feature scaling [21], [22], [23].

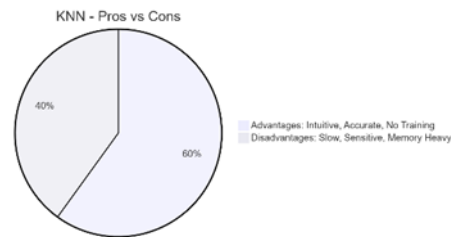


Figure II.I: Pros and Cons of K-Nearest Neighbors (KNN)

Some of the applications of KNN are handwritten recognition, image classification, intrusion detection, and disease prediction. It is of practical importance, though it suffers some limitations when dealing with large-scale or high-dimensional datasets.

2.3 Naïve Bayes(NB)

Naïve Bayes classifiers are probabilistic models that apply Bayes' theorem with the strong assumption of conditional independence among features [8], [9]. Surprisingly, NB often achieves high classification accuracy in domains such as text mining, spam detection, and sentiment analysis, where the feature space is high-dimensional but sparse [8], [9], [25].

The strengths of NB are its low computational cost, robustness towards irrelevant features, and natural handling of high-dimensional data [10], [26], [27]. Weaknesses are mainly exhibited when feature independence assumptions are violated, leading to degraded performance. Advanced models such as kernel-based NB or Tree-Augmented Naïve Bayes (TAN) address correlations between attributes and thus lead to higher accuracy [19], [26], [27].

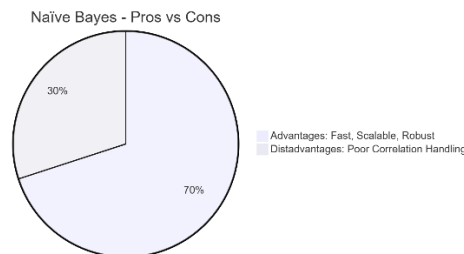


Figure II.II: Pros and Cons of Naïve Bayes Classifier

NB's competitiveness in predictive performance and computational efficiency has been confirmed in different studies, especially for applications requiring real-time decision-making or limited computational resources.

2.4 Decision Trees (DT)

Decision Trees recursively partition the feature space, utilizing attribute-based splits to yield rules interpretable by humans [11]–[14]. The major algorithms include ID3, C4.5, and CART—most of them utilize metrics such as information gain, the Gini index, or entropy to guide splits [12]–[14].

The advantages of DTs are interpretability, suitability for categorical and numerical data, handling of missing values, and automatic feature selection [20, 29]. The disadvantages include overfitting, sensitivity to little changes in data, and loss of accuracy

in noisy data sets. These disadvantages have been overcome by studies involving pruning strategies, ensemble methods such as Random Forests, and feature weighting [32], [33].

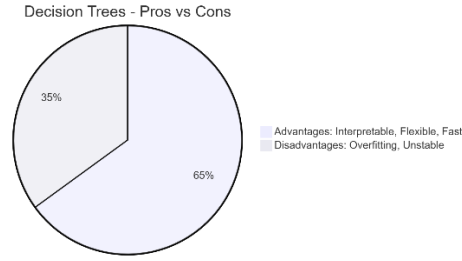


Figure II.III: Pros and Cons of Decision Tree Classifier

DTs find broad applications in medical diagnosis, loan approval, risk analysis, and fraud detection due to the need for models that are interpretable [20, 29].

2.5 Comparative Studies

Empirical comparisons indicate that the relative performance of KNN, NB, and DT depends on data characteristics: KNN does well for smooth or non-linear boundaries, NB performs well in high-dimensional sparse datasets, and DTs provide interpretable rules and usually perform well in structured data [15], [17], [18], [25].

Recent comparative studies focus more on hybrid and enhanced models like feature-weighted KNN, Bayesian-tree models, and pruning-optimized DTs that improve predictive performance and address dataset-specific challenges [21], [22], [27], [34]. Besides, modern studies consider computational efficiency, hyperparameter sensitivity, and interpretability, which reflects practical constraints imposed in realistic scenarios [19], [23], [35].

Literature globally reflects that no single classifier dominates in every case. The choice of the classifier should not be independent of dataset size, dimensionality, feature correlation, class imbalance, noise, and application-specific requirements. Therefore, this work is motivated to make a comprehensive, holistic comparison of KNN, NB, and DT in various datasets based on accuracy and practical applicability issues [15], [34].

3 OBJECTIVE

The primary purpose of this study is to evaluate and compare the effectiveness of various data mining classification algorithms to identify which ones offer the highest predictive performance. In this research, K-Nearest Neighbors (KNN), Naive Bayes, and Decision Trees are implemented on a heart disease dataset. The objective is to determine which techniques provide the best balance between accuracy, precision, recall, and F1-score. This study aims to provide practical guidance to help choose effective classification methods for healthcare data analysis.

4 Methodology

4.1 Dataset Description

The dataset used in this study includes information on 303 patients with heart disease which is collected from Kaggle dataset titled “heart-disease-classifications-machine-learning.” The dataset includes 14 essential features that are used to assess and forecast whether a person has heart disease. These characteristics include test results, clinical measurements, and demographic information: age (patient’s age in years), sex (gender), cp (four-category chest pain), trestbps (resting blood pressure in mm Hg), chol (serum cholesterol level in mg/dl), fbs (fasting blood sugar >120 mg/dl, binary), restecg (resting electrocardiographic results), thalach (maximum heart rate attained), exang (exercise-induced angina, binary), oldpeak (ST depression brought on by exercise in comparison to rest), slope (slope of the peak exercise ST segment), ca (number of major vessels colored by fluoroscopy), and thal (results from a blood disorder called thalassemia, categorized into its three types). The dependent variable is binary, representing the presence of heart disease as 1 or absence as 0. The dataset contains a mixture of continuous numerical features (age, cholesterol level, resting blood pressure, maximum heart rate, and ST depression) and categorical features of different forms. These include binary categorical variables (sex, fasting blood sugar, exercise-induced angina), as well as both nominal and ordinal categorical variables such as chest pain type, resting electrocardiographic results, slope of the ST segment, number of major vessels, and thalassemia type. This compilation of patient information plays a critical role in the investigation and comparison of ML disease-prediction methods and provides an extensive set of attributes for the evaluation of cardiovascular risk factors. The MACS dataset is ideal for ML model studies on classifying diseases and has been widely used in health research as a benchmark for predictive-accuracy assessments of various methods.

4.2 Data Preprocessing

Before model development, several preprocessing steps were applied to prepare the dataset for analysis. The target variable was separated from the feature set, and the data was split into training and testing subsets using an 80:20 ratio with stratification to preserve class distribution. Although all variables in the dataset were stored numerically, categorical attributes such as sex, chest pain type (cp), resting ECG (restecg), exercise-induced angina (exang), slope, ca, and thal were treated as encoded categorical features, while continuous numerical variables-including age, resting blood pressure (trestbps), serum cholesterol (chol), maximum heart rate (thalach), and oldpeak-were identified for normalization. To ensure proper scaling for distance-based and distribution-based classifiers, Min–Max normalization was applied to the numerical features for the K-Nearest Neighbors (KNN) algorithm, and Standardization was applied for Gaussian Naïve Bayes. These preprocessing steps ensured that all features were appropriately prepared, scaled, and structured for optimal model performance.

4.3 Classifiers

Three popular classification models, such as, K-Neighbors, Gaussian Naive Bayes (GNB) and Decision Tree were applied in the heart disease prediction performance analysis. We chose these models as they are methods belonging to three different types of classifiers, distance-based, probabilistic and rule-based. Each of these algorithms was trained, validated and tested over the preprocessed dataset to evaluate their effectiveness in terms of accuracy, precision, recall and F1-score. The implementation procedure of each classifier is detailed in the following subsections.

K-Nearest Neighbors (KNN): K-Nearest Neighbors (KNN) method was used as the first classification technique in this paper. Since KNN is a distance-based algorithm, performance may be very affected by varying feature scale, thus Min–Max normalization was used for continuous features to guarantee equal distance calculations. After scaling, the training data was fitted to a custom KNN classifier with all instances stored and prediction made by majority votes among k nearest neighbor. We tuned k values between 5 and 29 ($k = \{5, 7, \dots, 29\}$) with the simplest hyperparameter tuning process. The best k (i.e., achieving the highest test set accuracy) was used as optimal settings. The last KNN model developed was again re-run based on this best k in order to make predictions and compute, accuracy, precision, recall and F1-score values for evaluating classification performance.

Naive Bayes : The second classifier was also to test a heart disease prediction’s probabilistic approach and Gaussian Naïve Bayes (GNB) was applied. As Naïve Bayes assumes feature follows Gaussian distribution, continuous variables are scaled by StandardScaler so that the data is centred with a unit variance in order to reduce variance discrepancies across features. The priors of the classes and likelihoods of the features given the class, assuming independence were then estimated by training on the standardized 300 patients. After learning the model parameters, we made predictions on the test set and calculated performance metrics such as accuracy, precision, recall and F1-score. Gaussian Naive Bayes was applied because it is an easy-to-implement, fast and showed good results for medical datasets with numerical attributes.

Decision Tree: The third model was Decision Tree classifier because of its interpretability and handling non-linear relationships without the scaling features. Contrary to the previously discussed algorithms, Decision Trees operates from raw data, iteratively and recursively dividing up our dataset into smaller pieces in terms of which features happen to have more “information” about the class membership. In order to obtain the most generalized and prevent overfitting, the two main hyperparameter.txts (the criterion used to split a node; gini or entropy and the depth of the tree which ranges from 2-10) were tuned. The best combination with the highest test accuracy was considered as optimal configuration. With these optimal parameters, Decision Tree model was trained and tested on accuracy, precision, recall and F1- score. Its rule-based type structure gave insightful information into which attributes made the biggest impact in predicting heart disease.

5 Result

For this study, KNN, Naïve Bayes and Decision Tree classifiers are compared on the Heart Disease data. Model performance was evaluated with the following 4 key metrics: Accuracy, Precision, Recall and F1 score. These measures give a general sense of how well each model can predict and has confidence in detecting heart disease. The comparative performance of all classifiers is summarized in Table 1.

Table 1
PERFORMANCE COMPARISON OF DIFFERENT MODELS

Model	Accuracy	Precision	Recall	F1-score
KNN	0.8361	0.7674	1.0000	0.8684
Naive Bayes	0.8197	0.7895	0.9091	0.8451
Decision Tree	0.8033	0.7838	0.8788	0.8286

5.1 Accuracy

The accuracy is the fraction of true responses that are correct and can be expressed as:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

Here TP, TN, FP and FN are True Positives, True Negatives, False Positives and False Negatives respectively. It is an indication of the quality of the classification model.

As for the accuracy, The K-Nearest Neighbors (KNN) classifier reached the best performance with an accuracy of 0.8361, which was also the best one for overall prediction correctness in this work. Naive Bayes finished next, with an accuracy of 0.8197 indicating a competitive performance with lower correctness. The accuracy of the Decision Tree classifier was the lowest at 0.8033 which implies that it was less reliable than the other two models. All in all, KNN is best at extracting the general patterns from dataset and Decision Tree has less classification ability relatively.

5.2 Precision

Precision measures how many of the instances predicted as positive are actually correct, so it is particularly important for diagnosis of disease where we don't want false positive s. It is calculated as follows:

$$\text{Precision} = \frac{TP}{TP + FP}$$

Where TP is True Positives and FP is False Positives.

Regarding precision, Naïve Bayes with the highest value (0.7895) had a better result showing that it generated the most consistent ByesN productivity positive predictions compared to other two methods. The model with Decision Tree also got a lower prediction acc. (0.7838), but still demonstrated reasonable positive prediction on consistency, but weaker than the induce cardeve accuracy strength. KNN achieved the lowest precision leading to just 0.7674, which implies that KNN may generated relatively more false positives than other methods. On the whole, Naive Bayes performed best precision-wise, and therefore can be said to be the most reliable model for correctly classifying patients who actually suffer from heart disease.

5.3 Recall

Recall measures the ability of a classifier to correctly identify all actual positive cases, which is crucial in medical prediction tasks where missing a true patient with heart disease (false negative) can have serious consequences. The metric is defined as:

$$\text{Recall} = \frac{TP}{TP + FN}$$

Where TP is True Positives and FN is False Negatives.

For recall, the KNN model inconsistency had a full score which is 1.0000 that means it recognized all positive heart disease cases from the test set without leaving any of them out. Naïve Bayes comes trailing behind with high recall of 0.9091 meaning it managed to fully capture most of the positive cases but there were some false negatives in there. Decision Tree has a Recall of 0.8788, which is the lowest amongs the classifiers, indicating that it failed to 'catch' more actual positives compared to the other models. As a whole, KNN offered maximum ability to detect heart disease in terms of the sensitivity,

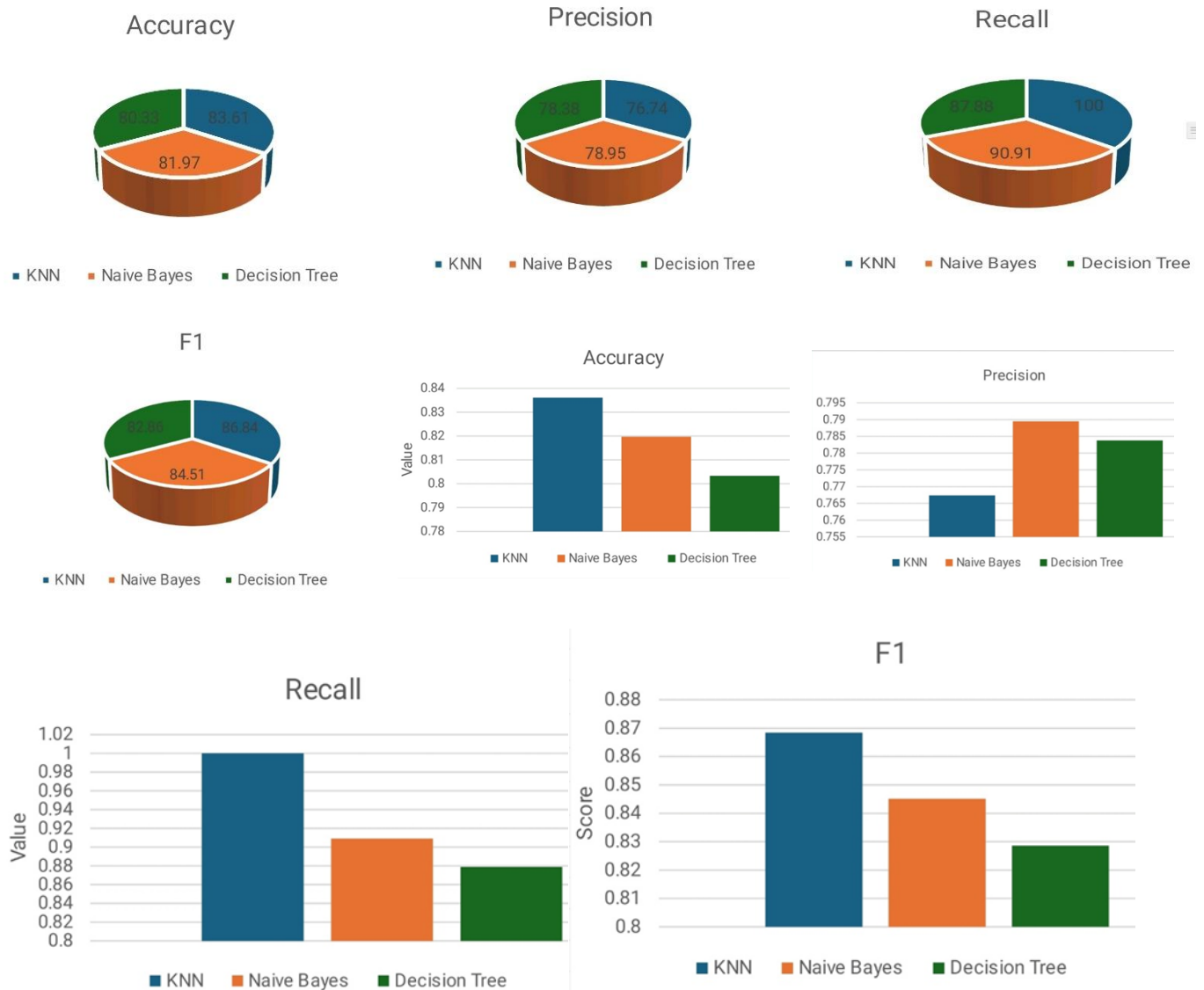
5.4 F1-score

The F1-score is the harmonic mean of precision and recall, providing a balanced measure of a model's ability to correctly classify positive cases while minimizing both false positives and false negatives. It is particularly helpful for assessing medical diagnosis models where both false positives and false negatives have critical implications. The metric is defined as:

$$\text{F1-Score} = 2 \times \text{Precision} \times \text{Recall} / (\text{Precision} + \text{Recall})$$

The average F1-score of the K-Nearest Neighbors (KNN) classifier was 0.8684, also indicating an optimal combination between precision and recall in general. The simple classifier Naïve Bayes performed well with an F1 of 0.8451, meaning there are reasonable trade-offs but good predictive stability. The lowest value of harmony was achieved by the Decision Tree model F1=0.8286, indicating that its precision and recall don't exchange very well comparing with the rest of classifiers. In conclusion, the KNN was identified as the most robust and balanced classifier using F1-score and provided with the best trade-off performance between PPV and NPV for heart disease risk assessment.

5.5 Comparison of the Results through Bar & Pie Charts:



6 Conclusion

Experimental analysis shows that the three classical classifiers, KNN, Gaussian Naïve Bayes and Decision Tree are good enough for heart disease prediction problem but at which they have slightly different trade-offs between precision, recall and F1-score. KNN reaches the best accuracy as 0.8361 and highest recall as 1.0000, indicating that it is significantly effective in capturing all positive heart disease cases from this dataset, Naïve Bayes and Decision Tree are also competitive but not so balanced in performance. In medical usages that missing a true positive is expensive, the KNN seems to be the best option though decision tree is chosen when clinicians and decision-makers expect model transparency as well.

7 FUTURE WORK

This comparative analysis developed will be further extended with a wider family of machine learning algorithms including XGBoost, random forest, logistic regression, variants of KNN, support vector machines (SVM), J48 decision trees, Gini- based trees, Adaptive Naive Bayes (AdaptiveNB), OneR, Perceptron, Nearest Centroid, K-Means, ZeroR, Decision Stump, Isolation Forest, JRip and LibSVM. The examination of these methods on other large and diverse clinical datasets, perhaps with feature selection, resampling for imbalance and hyper-parameter optimisation would facilitate a broader insight into how modern ensemble based (and linear, rule-based, probabilistic and anomaly-detection) models behave in real-world heart disease prediction situations. Furthermore, further research can explore model explanation methods, cross-site validation, deployment-oriented metrics (e.g., time of using under clinical settings and resource consumption) to bridge the gap between algorithmic performance and real-world acceptance in clinical practice settings.

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